From CA to Agent-Based Models

The Schelling Segregation Model & Introduction to Mesa

From Conway to LangGraph • Week 3

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Today's Roadmap

I **From CA to ABM**
The conceptual leap: when cells become agents

II **The Schelling Model**
Mild preferences, extreme segregation

III **Mathematical Framework**
Order parameters, phase transitions, Ising analogy

IV **Introduction to Mesa**
Agent-based modeling in Python

V **Implementation Lab**
Building Schelling in Mesa, parameter sweeps

PART I

From CA to ABM

When cells become agents with desires

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Cellular Automata — What We Know

Cellular Automaton

Lattice $L = Z^d$

Finite state set
$S = \{0, 1, \ldots, k-1\}$

Neighborhood
$N(i) \subset L$

Local rule
$f : S|N| \rightarrow S$

Synchronous update

Key Properties

- ✓ Deterministic evolution
- ✓ Homogeneous rules (same for all cells)
- ✓ No internal state beyond cell value
- ✓ No decision-making or utility
- ✓ Cells don't move
- ✓ Global behavior from local rules

What Makes an Agent?

An agent is an entity that:

Perceives

Observes its local environment (neighborhood)

Evaluates

Computes a utility or satisfaction function

Decides

Chooses an action based on evaluation

Acts

Moves, changes state, interacts with others

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CA vs Agent-Based Model

Property	Cellular Automaton	Agent-Based Model
Entity	Cell (fixed position)	Agent (autonomous)
State	Cell value $\in S$	Internal state vector
Movement	None — cells are static	Agents can relocate
Rules	Deterministic function f	Utility-based decisions
Update	Synchronous (all at once)	Often asynchronous
Interaction	Via shared neighborhood	Can be non-local
Heterogeneity	Homogeneous rules	Agents can differ

The Bridge: GoL as Proto-ABM

Conway's Game of Life is already halfway there:

GoL as CA

Cells on a grid
Birth/survival rules
Deterministic update
No concept of "wanting"
Fixed topology

→

GoL reinterpreted as ABM

"Organisms" on a grid
Local "social" interactions
Emergent structures (gliders!)
Self-replication, movement
But still no preferences...

Key question: *What happens when we give cells preferences and the ability to move?*

PART II

The Schelling Model

Micro-motives and macro-behavior

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Thomas Schelling (1921–2016)

Nobel Prize in Economics (2005)

"for having enhanced our understanding of conflict and cooperation through game-theory analysis"

Micromotives and Macrobehavior (1978)

Original insight: Schelling used a physical checkerboard with coins to show that even mild individual preferences for same-type neighbors inevitably produce large-scale residential segregation.

This was one of the first agent-based models — predating computers' widespread use for social simulation.

"There is no individual intent to segregate."

The model shows how macro-patterns emerge from micro-motives that are individually innocent.

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Schelling Model — Formal Definition

Grid

$L \subset Z^2$ square lattice of size $N \times N$

State

Each cell $\sigma_i \in \{ \varnothing, A, B \}$ (empty, type A, or type B)

Tolerance

Parameter $T \in [0, 1]$ — minimum fraction of same-type neighbors required

Agent i is **happy** if $(\text{same-type neighbors}) / (\text{total neighbors}) \geq T$

Dynamics

1. Pick a random unhappy agent
2. Move it to a random empty cell
3. Repeat until no unhappy agents remain (or max steps)

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The Surprising Result

Even with mild preferences, segregation is inevitable

T = 0.30

"I want at least 30% like me"

Do IT !!!

[simulation snapshot]

Strong segregation

T = 0.50

"I want at least half like me"

[simulation snapshot]

Near-total segregation

T = 0.75

"I want a clear majority like me"

[simulation snapshot]

Unstable / frozen clusters

Why Is This Surprising?

The Paradox of Unintended Consequences

At T = 1/3, each agent is perfectly happy in a mixed neighborhood (e.g., 3 same + 5 different out of 8 neighbors). Nobody wants segregation. Yet the collective dynamics produce it — a textbook example of emergence.

Physics Analogy

This is analogous to spontaneous symmetry breaking in the Ising model. The Hamiltonian is Z_2-symmetric, but the ground state is not. Similarly, Schelling's rules are symmetric between A and B, but equilibria are segregated.

Computational Perspective

The model has many Nash equilibria, most of which are segregated. The random dynamics act as a "selection mechanism" that drives the system toward segregated attractors, even from well-mixed initial conditions.

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PART III

Mathematical Framework

Order parameters, phase transitions, and the Ising analogy

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Utility Function

Agent Utility

For agent i of type $\tau \in \{A, B\}$ with neighbors $N(i)$:

$$u_i = |\{ j \in N(i) : \sigma_j = \tau \}| \;/\; |\{ j \in N(i) : \sigma_j \neq \varnothing \}|$$

Decision Rule

Agent is **happy** if $u_i \geq T$ and **unhappy** if $u_i < T$

Note:

Unlike physics models where energy is minimized continuously, here utility is binary (happy/unhappy). This discrete jump makes the dynamics non-trivial — there's no gradient to follow, agents simply flee unsatisfactory positions.

Order Parameter: Segregation Index

Global Segregation Index

Average fraction of same-type neighbors over all agents:

$$S = (1/N) \Sigma_i u_i$$

Interpretation

$S \approx 0.5$	Well-mixed (random)
$S \to 1.0$	Fully segregated
$S \approx T$	Near critical threshold

Ising Analogy

S plays the same role as magnetization |m| in the Ising model:

$m = (1/N) \Sigma_i s_i$

Both measure the degree of ordering. Tolerance T maps (inversely) onto temperature: low T ≈ high temperature (disordered), high T ≈ low temperature (ordered).

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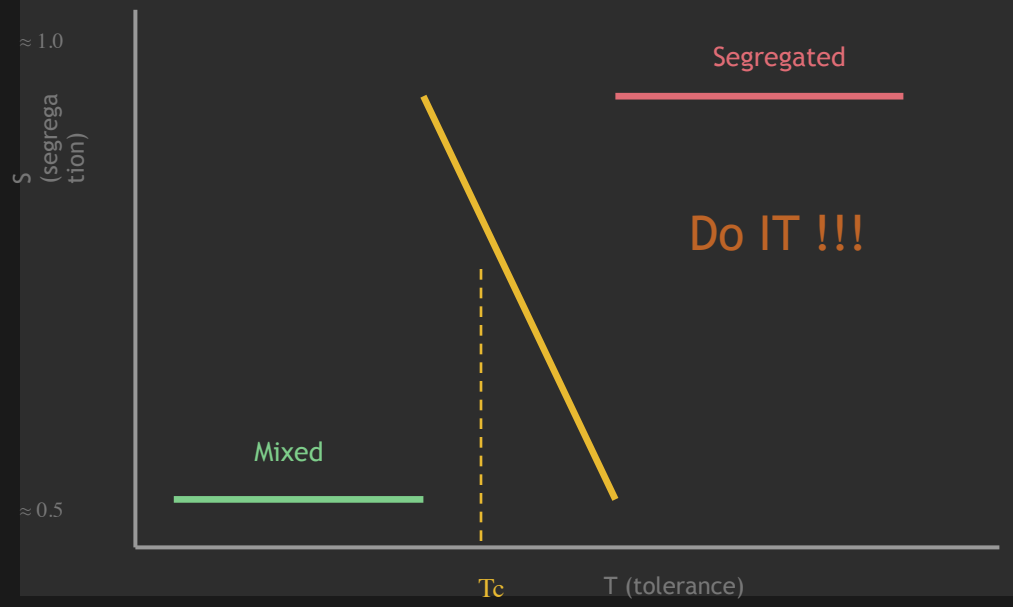
Phase Transition in the Schelling Model

Phase Diagram (schematic)



Key Features

Sharp transition at critical tolerance Tc ≈ 0.30–0.35 (depends on density, grid size)

Below Tc: random initial conditions → random equilibrium

Above Tc: random → segregated (symmetry breaking)

Finite-size effects: transition sharpens with N → ∞

Analogous to ferromagnetic transition in Ising

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Mean-Field Approximation

Simplifying Assumption

In mean-field theory, we replace local neighborhood statistics with global averages. If the fraction of type A agents is ρ and agents are randomly placed:

$$P(\text{agent A is happy}) = P(\text{Binomial}(k, \varrho) / k \geq T)$$

Mean-Field Predictions

For Moore neighborhood (k = 8) with ρ = 0.5:

T = 0.25: P(happy) ≈ 0.96
T = 0.375: P(happy) ≈ 0.86
T = 0.50: P(happy) ≈ 0.64
T = 0.625: P(happy) ≈ 0.36

Where Mean-Field Fails

Mean-field predicts a smooth crossover, but simulations show a sharp transition. The key missing ingredient: spatial correlations. Once clusters form, they self-reinforce — neighboring unhappy agents move together, creating feedback loops that mean-field cannot capture.

PART III-A

Mean-Field Deep Dive

Exact calculations, where they work, and why they fail

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Mean-Field: The Core Idea

The Assumption

Replace local neighborhood statistics with global averages. Each agent "sees" a random sample of the population, ignoring spatial correlations.

If fraction of type A agents is ρ:

Each neighbor is independently type A with probability ρ, type B with (1 − ρ)

This is exact for:

- ∞-dimensional lattice
- Complete graph (all-to-all)
- Fully random placement

Formal Statement

For an agent of type A with k neighbors:

```
X ~ Binomial(k, ρ)
```

where X = number of same-type neighbors.

Agent is happy iff $X/k \geq T$, i.e. $X \geq \lceil kT \rceil$:

```
P(happy) = 1 − I_ρ(⌈kT⌉, k−⌈kT⌉+1)
```

where I_ρ is the regularized incomplete beta function (CDF of the binomial).

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Mean-Field Predictions

Moore neighborhood: k = 8 neighbors, ρ = 0.5 (equal populations)

T	⌈kT⌉	P(happy) MF	P(happy) Sim	⟨S⟩ MF	⟨S⟩ Sim	Verdict
0.125	1	0.996	≈ 0.99	0.50	≈ 0.50	✓ Good
0.250	2	0.965	≈ 0.96	0.50	≈ 0.52	✓ Good
0.375	3	0.855	≈ 0.85	0.50	≈ 0.72	✗ Fails!
0.500	4	0.637	≈ 0.55	0.50	≈ 0.85	✗ Fails!
0.625	5	0.363	≈ 0.30	0.50	≈ 0.90	✗ Fails!
0.750	6	0.145	≈ 0.12	0.50	≈ 0.93	✗ Fails!

Key insight: MF predicts P(happy) reasonably well, but completely misses the segregation index ⟨S⟩. Why? Because P(happy) is a local property, while ⟨S⟩ depends on spatial structure that MF ignores.

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Self-Consistent Mean-Field Equation

The Naive vs Self-Consistent Approach

Naive MF:
Assume agents are randomly placed → compute P(happy) → done.

Self-consistent MF:
When unhappy agents move, they change local density. Let ρ^* be the effective local density after relaxation:

$$\rho^* = \rho_0 + \delta\rho(T, \rho^*) \quad \text{where } \delta\rho \text{ encodes segregation feedback}$$

The Feedback Loop

1. Agent A unhappy → moves away
2. Departure reduces diversity for old neighbors
3. A arrives → may displace others at new spot
4. Cascade of moves → cluster growth

Ising Analogy

Weiss MF for Ising:

$$m = \tanh(\beta J z m)$$

Predicts phase transition correctly but gets Tc wrong and misses critical exponents.
Same story for Schelling!

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Why Mean-Field Fails

The three missing ingredients

Spatial Correlations

MF assumes:

$$\langle \sigma_i \sigma_j \rangle = \langle \sigma_i \rangle \langle \sigma_j \rangle$$

Reality:

Once a cluster forms, neighboring agents are strongly correlated. The 2-point correlation function $C(r) = \langle \sigma_0 \sigma_r \rangle - \langle \sigma \rangle^2$ decays as a power law near Tc, not exponentially.

Interface Effects

MF assumes:

Agents in a uniform bath

Reality:

Unhappy agents concentrate at cluster boundaries. Interface tension drives coarsening: small clusters get absorbed by large ones, following the Allen-Cahn equation — domain size grows as $L(t) \sim t^{(1/2)}$.

Positive Feedback

MF assumes:

Independent agent moves

Reality:

Each move makes same-type neighbors happier but different-type neighbors sadder. This is a positive feedback that amplifies fluctuations — exactly the mechanism MF cannot capture (it averages them away).

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Quantifying the Failure

Correlation Function

Define spin variable: $s_i = +1$ (type A), -1 (type B)

$$C(r) = \langle s_i \, s_j \rangle - \langle s \rangle^2 \quad \text{for } |i-j| = r$$

MF predicts: $C(r) = 0$ for all $r > 0$
Simulation: $C(r) \sim r^{(-\eta)}$ near Tc
with $\eta \approx 0.25$ (Ising-like!)

Correlation Length ξ

Away from Tc, correlations decay:

$$C(r) \sim \exp(-r/\xi)$$

MF is valid when $\xi \ll$ lattice spacing (neighbors are uncorrelated).

Near Tc: $\xi \to \infty$ (diverges!)
→ MF fails catastrophically

The Ginzburg Criterion

MF is valid when fluctuations within a correlation volume are small compared to the mean:

$$\langle (\delta S)^2 \rangle_\xi \, / \, \langle S \rangle^2 \ll 1 \quad \to \quad |T - T_c| \gg G_i$$

The Ginzburg number Gi depends on dimensionality d. For d = 2 (our grid), Gi is large — MF fails over a wide region around Tc. This is why the simulated transition is so much sharper than MF predicts.

Beyond Mean-Field

Better analytical approaches to the Schelling model

1 Pair Approximation

Track 2-point correlations $\langle s_i s_j \rangle$ explicitly. Close the hierarchy at the triplet level. Gives much better Tc estimate but still misses critical exponents.

2 Effective Hamiltonian

Map Schelling onto an Ising-like model with effective couplings J(T) and external field h(T). Dall'Asta et al. (2008) showed this recovers the correct universality class for symmetric populations.

3 Interface / Coarsening Theory

After initial segregation, domain growth follows curvature-driven coarsening. Average domain size L(t) ~ t^(1/2) matches Model A dynamics (non-conserved order parameter) in the Hohenberg-Halperin classification.

4 Monte Carlo RG

Finite-size scaling analysis of simulations. Gauvin et al. (2009) measured critical exponents $\beta \approx 0.125$, $\gamma \approx 1.75$ — consistent with the 2D Ising universality class. A remarkable confirmation that Schelling $\in$ Ising!

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Mean-Field: A Physicist's First Tool

What MF Gets Right ✓

- Existence of a phase transition
- Qualitative shape of S(T): smooth increase above a threshold
- Local happiness P(happy) reasonably accurate
- Correct prediction that $\rho = 0.5$ is most stable

What MF Gets Wrong ✗

- Segregation index (off by up to 80%!)
- Critical tolerance Tc (predicts smooth crossover, not sharp transition)
- Critical exponents (predicts MF exponents $\beta=1/2$, not Ising $\beta\approx1/8$)
- Coarsening dynamics and relaxation times

The Lesson for Physicists

Mean-field is the physicist's "Hello World" — the first thing you try, and it tells you whether a phenomenon is interesting. If MF predicts no transition, there probably isn't one. If MF predicts a transition but gets the details wrong, congratulations: you've found a system where **correlations and fluctuations matter** — and that's where the real physics lives.

→ *Exercise 3 (Mean-Field vs Simulation) now has much richer structure to explore!*

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PART IV

Introduction to Mesa

Agent-based modeling in Python

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Mesa: ABM Framework for Python

What is Mesa?

An open-source Python framework for agent-based modeling. Think of it as "NumPy for agents" — it provides the building blocks so you can focus on the model, not the infrastructure.

Why Mesa for This Course?

Mesa's architecture (Agent → Model → step) maps directly onto LangGraph's architecture (Node → Graph → invoke) that you'll use in Weeks 9–12. Learning Mesa now builds the right mental model.

```
pip install mesa
```

github.com/projectmesa/mesa

3.2k+
GitHub stars

700+
Contributors

v3.x
Current version

Mesa Architecture

Agent

Individual entity with state and step() method. Each agent perceives, evaluates, decides.

```
class MyAgent(mesa.Agent):
    def step(self):
        # perceive, decide, act
```

Model

Container: holds agents, grid, scheduler, and global parameters.

```
class MyModel(mesa.Model):
    def __init__(self, N, ...):
        # create grid, agents
```

Space

Grid or network topology. Agents query neighbors, move between cells.

```
grid = mesa.space.SingleGrid(
    width, height, torus=True)
```

DataCollector

Records model-level and agent-level data at each step for analysis.

```
dc = mesa.DataCollector(
    model_reporters={"seg": fn})
```

Mesa → LangGraph: The Conceptual Bridge

The same architecture pattern, two decades apart:

Mesa (ABM)	↔	LangGraph (AI Agents)
Agent.step()	→	Node function
Model (container)	→	StateGraph
Scheduler (order)	→	Edges / routing
Grid (topology)	→	State (shared memory)
DataCollector	→	Checkpointer
agent.model.grid	→	state dict

Master Mesa now → LangGraph will feel natural in Week 9

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PART V

Implementation Lab

Building the Schelling Model in Mesa

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SchellingAgent Class

```python
class SchellingAgent(mesa.Agent):
    """An agent in the Schelling segregation model."""

    def __init__(self, model, agent_type):
        super().__init__(model)
        self.type = agent_type  # 0 or 1

    def step(self):
        neighbors = self.model.grid.get_neighbors(
            self.pos, moore=True, include_center=False
        )
        similar = sum(
            1 for n in neighbors if n.type == self.type
        )
        # Calculate utility
        if len(neighbors) > 0:
            self.utility = similar / len(neighbors)
        else:
            self.utility = 1.0

        # Decision: move if unhappy
        if self.utility < self.model.tolerance:
            self.model.grid.move_to_empty(self)
```

Perceive

get_neighbors() queries the Moore neighborhood on the grid

Evaluate

Compute utility = fraction of same-type neighbors

Decide & Act

If utility < T → move to random empty cell

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SchellingModel Class

```python
class SchellingModel(mesa.Model):
    def __init__(self, width=50, height=50,
                 density=0.8, tolerance=0.375,
                 minority_frac=0.5, seed=None):
        super().__init__(seed=seed)
        self.tolerance = tolerance
        self.grid = mesa.space.SingleGrid(
            width, height, torus=True
        )
        # Place agents randomly
        for cell in self.grid.coord_iter():
            x, y = cell[1], cell[2]
            if self.random.random() < density:
                agent_type = (
                    0 if self.random.random() < minority_frac
                    else 1
                )
                agent = SchellingAgent(self, agent_type)
                self.grid.place_agent(agent, (x, y))

        self.datacollector = mesa.DataCollector(
            model_reporters={
                "segregation": compute_segregation,
                "happy_frac": compute_happy_fraction
            }
        )
```

Key Parameters

density

Fraction of cells occupied (0.8 = 80%)

tolerance

Threshold T for happiness

minority_frac

Ratio of type A to B

SingleGrid

At most one agent per cell

torus=True

Periodic boundaries

Running the Simulation

```
# Run a single simulation
model = SchellingModel(
    width=50, height=50,
    density=0.8, tolerance=0.375
)

for step in range(100):
    model.step()

# Collect data
data = model.datacollector.get_model_dataframe()
```

```
# Parameter sweep
from mesa.batchrunner import batch_run

results = batch_run(
    SchellingModel,
    parameters={
        "tolerance": np.arange(0.1, 0.8, 0.05),
        "density": [0.7, 0.8, 0.9],
    },
    iterations=10,
    max_steps=200,
)
```

What to Measure

Segregation index S(t)
Order parameter vs time

Happy fraction(t)
Convergence diagnostic

S(T) at equilibrium
Phase diagram

Relaxation time τ(T)
Critical slowing down

Cluster size distribution
Percolation analysis

S vs N at fixed T
Finite-size scaling

Visualizing the Dynamics

```python
import matplotlib.pyplot as plt
from matplotlib.colors import ListedColormap

def plot_schelling(model, ax=None):
    """Visualize current grid state."""
    if ax is None:
        fig, ax = plt.subplots(figsize=(8, 8))

    grid = np.zeros((model.grid.width, model.grid.height))
    for agent in model.agents:
        x, y = agent.pos
        grid[x][y] = 1 if agent.type == 0 else 2

    cmap = ListedColormap(["#1A1A1A", "#4A9ECC", "#E8B931"])
    ax.imshow(grid.T, cmap=cmap, interpolation="nearest")
    ax.set_xticks([])
    ax.set_yticks([])
    return ax
```

Color Scheme

- Empty cell
- Type A (steel blue)
- Type B (gold)

Matches course color palette!

Pro tip: Use matplotlib.animation.FuncAnimation to create step-by-step evolution movies. The visual impact of watching segregation emerge from a random initial condition is powerful.

PART VI

Exercises

Hands-on exploration of the Schelling model

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Week 3 Exercises

Exercise 1 — **Phase Diagram Construction** — 45 min • Core

Sweep T ∈ [0.1, 0.8] in steps of 0.05. For each T, run 10 simulations with different seeds. Plot S(T) with error bars. Identify the critical tolerance Tc.

Exercise 2 — **Finite-Size Scaling** — 60 min • Core

Repeat Exercise 1 for grid sizes N = 20, 30, 50, 80. Plot S(T, N) and study how the transition sharpens. Estimate the scaling exponent ν from the crossing point.

Exercise 3 — **Mean-Field vs Simulation** — 45 min • Core

Compute the mean-field prediction for P(happy) as a function of T using the binomial distribution. Compare with measured happy fraction from simulations. Where does mean-field break down?

Exercise 4 — **Asymmetric Populations** — 45 min • Advanced

Vary minority_frac from 0.1 to 0.5. How does the phase diagram change? Is the minority more or less segregated? Connect to percolation theory.

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Key Takeaways

- ABMs extend CA by giving cells agency: internal state, utility, decisions, movement
- Schelling's model: mild individual preferences → extreme collective segregation (emergence!)
- The model exhibits a sharp phase transition at critical tolerance $T_c \approx 0.30$–0.35
- Mean-field theory captures qualitative behavior but misses the sharpness of the transition
- Mesa provides the architectural pattern (Agent → Model → step) that scales to LangGraph

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Next Week: Self-Organized Criticality

Week 4 Preview

From equilibrium phase transitions to systems that self-organize to criticality — no parameter tuning needed.

- **Bak-Tang-Wiesenfeld Sandpile**
  The canonical SOC model: avalanches, power laws, 1/f noise
- **Power-Law Distributions**
  Heavy tails, scale invariance, Zipf's law
- **Universality & Scaling**
  Critical exponents, fractal dimension of avalanches
- **SOC in Nature**
  Earthquakes, forest fires, neural activity, financial markets

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Questions?

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